

E-Commerce Customer Analytics

RFM Segmentation, Market Basket Rules, and Revenue Forecasting

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ABSTRACT

This project develops an end-to-end data mining pipeline for e-commerce customer analytics using the Online Retail II transaction dataset, which records invoice-level purchases for a UK-based non-store retailer between 2009 and 2011. The central goal is to move from raw transactional logs to actionable structures for marketing, merchandising, and revenue planning. The pipeline addresses challenging data quality issues – including returns, cancellations, non-product codes, missing customer identifiers, and heavy-tailed sales distributions – through systematic cleaning, filtering, and aggregation.

On the customer side, constructed a behavior-only feature space via Recency–Frequency–Monetary (RFM) statistics and apply K-Means clustering, with the number of clusters selected using silhouette-based model selection. The resulting four segments exhibit clearly differentiated purchasing patterns and monetary contributions, supporting interpretations such as “Champions,” “Loyal,” “New/Potential,” and “At-Risk” customers and enabling segment-specific retention and upsell strategies. At the basket level, transformed invoices into sparse binary baskets and employ the Apriori algorithm to mine frequent itemsets and association rules. High-lift rules reveal coherent product constellations – such as multi-color tea sets and composite playhouse kits – that can be directly translated into bundle design, cross-sell placements, and stock coordination.

For temporal dynamics, aggregated UK revenue to a daily time series and compare multiple forecasting models on a held-out test window in late 2011. A simple seasonal ARIMA model serves as the baseline, against which a tuned SARIMA specification and a Prophet model achieve substantially lower RMSE and MAE, while capturing weekly seasonality and holiday peaks. Taken together, the segmentation, association mining, and forecasting components demonstrate how classical data mining techniques, applied carefully to a single historical dataset, can yield a coherent and operationally relevant view of e-commerce customers, products, and revenue trajectories.

Code and resources:

https://github.com/neehanthereddy/e-commerce_analytics_pipeline

KEYWORDS

E-commerce analytics, Customer segmentation, RFM (Recency–Frequency–Monetary), K-Means, Market basket analysis, Association rule mining, Apriori algorithm, Sales forecasting, SARIMA, Prophet

1 Introduction

Transactional data from e-commerce platforms are often abundant but under-exploited. Businesses routinely compute simple aggregates such as total monthly sales or the list of top-selling products, yet strategic decisions about customer retention, cross-selling, and inventory are still frequently driven by ad-hoc intuition. This project takes a step toward more systematic decision support by designing a coherent data-mining pipeline for a real transaction dataset.

The empirical basis is the Online Retail II dataset, which records invoices from a UK-based, non-store retailer specializing in all-occasion giftware. Each transaction includes an invoice identifier, item code and description, quantity, timestamp, unit price, customer identifier, and country. The observation window runs from 1 December 2009 to 9 December 2011, covering almost two years of sales activity. The dataset is attractive not only because it is widely used as a benchmark, but also because its structure closely resembles the operational tables of many small and medium-sized retailers.

Working with this dataset presents several non-trivial challenges. First, the raw data contain numerous cancellations, returns, and administrative records such as postage fees or bank charges; these rows are encoded via negative quantities, special invoice prefixes, or non-product stock codes. Second, a substantial fraction of transactions lacks a customer identifier, which complicates any attempt to construct longitudinal customer-level features. Third, the numerical distributions are highly skewed: both order quantities and revenue exhibit heavy tails driven by a small number of large orders, and customer-level frequency and monetary value are similarly concentrated in a handful of buyers. Finally, there is a strong seasonal component due to holiday sales, which complicates time-series modeling by introducing sharp spikes.

The project is organized around three application questions that are both realistic and analytically rich. The first is how to divide the customer base into a small number of interpretable behavioral segments that can support targeted marketing strategies. The second is how to discover which products tend to be purchased together so that the retailer can design bundles, cross-sell recommendations, and more informed merchandising. The third is how to forecast short-term sales, at least at the aggregate country level, to support staffing, inventory, and cash-flow planning.

To address these questions, the project implements a pipeline with three principal modules. A customer segmentation module computes RFM features at the customer level, normalizes them, and applies K-Means clustering. A market basket module converts invoice-level data to a binary invoice-product matrix and mines frequent itemsets and association rules using the Apriori algorithm. A forecasting module aggregates UK revenue to a daily time series and fits and compares several models, focusing on SARIMA and Prophet. The remainder of the report follows the required structure: Section 2 details the solution and modeling choices, Section 3 presents empirical results, Section 4 discusses limitations and potential extensions, and Section 5 concludes.

2 Solution

2.1 Data Preparation and Cleaning

The analysis is implemented in Python within a single Jupyter notebook, using Pandas and NumPy for data manipulation, Matplotlib and Plotly for visualization, scikit-learn for clustering, mlxtend for association rules, and statsmodels and Prophet for time-series models.

The original dataset contains 541,910 rows. An initial diagnostic step prints the schema, confirms data types, and reveals that the invoice and stock code fields are strings, while quantity and unit price are numeric. CustomerID is present in only about three quarters of the rows. Several quality issues are apparent: there are cancellations encoded as invoices whose identifiers begin with the letter “C”, there are stock codes that clearly correspond to non-product administrative entries such as postage charges or fees, some unit prices are zero or negative, and many rows are missing a customer identifier.

Cleaning proceeds in a sequence of filters that gradually enforce the structure required for downstream analysis. First, invoice identifiers are cast to strings and only those matching a six-digit numeric pattern are retained. This step automatically excludes cancellations and a small set of special invoices with alphabetic prefixes. Second, a curated list of non-product stock codes is constructed by inspecting the frequency and descriptions of stock codes; entries such as POST, DOT, D, C2, M, m, BANK CHARGES, AMAZONFEE, CRUK, B, and S are removed because they correspond to postage, discounts, carriage, bad debt, or fee adjustments rather than physical items. Additionally, a mask ensures that surviving stock codes either match a five-digit numeric pattern typical of ordinary products or belong to a small

whitelist of exceptional but legitimate codes such as PADS. Third, any rows with missing CustomerID are dropped for the purposes of customer-level analysis, and transactions with non-positive unit prices are discarded to avoid free samples and obvious data entry errors.

After these steps, the cleaned dataset consists of 363,090 rows, meaning roughly one third of the original entries are removed. The cleaned frame has no missing values in InvoiceNo, StockCode, Description, Quantity, UnitPrice, CustomerID, or Country. A new revenue column, often called TotalPrice in the notebook, is computed as the product of Quantity and UnitPrice, which serves as the basis for both monetary features and aggregated revenue time series. This cleaned data frame is the common input for all subsequent modules.

#	Column	Non-Null Count	Dtype
0	InvoiceNo	541910 non-null	object
1	StockCode	541910 non-null	object
2	Description	540456 non-null	object
3	Quantity	541910 non-null	int64
4	InvoiceDate	541910 non-null	object
5	UnitPrice	541910 non-null	float64
6	CustomerID	406838 non-null	float64
7	Country	541910 non-null	object

dtypes: float64(2), int64(1), object(5)
memory usage: 33.1+ MB

#	Column	Non-Null Count	Dtype
0	InvoiceNo	363090 non-null	object
1	StockCode	363090 non-null	object
2	Description	363090 non-null	object
3	Quantity	363090 non-null	int64
4	InvoiceDate	363090 non-null	datetime64[ns]
5	UnitPrice	363090 non-null	float64
6	CustomerID	363090 non-null	float64
7	Country	363090 non-null	object

dtypes: datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 24.9+ MB

Figure 1: a summary of the data.info() output comparing the raw and cleaned datasets

2.2 Customer Segmentation via RFM and K-Means

Customer segmentation begins by aggregating the cleaned transactions to the customer level. The analysis adopts the standard RFM framework. A reference date is chosen as one day after the last observed invoice date in the dataset. For each distinct CustomerID, recency is calculated as the number of days between the reference date and the most recent invoice; frequency is computed as the number of distinct invoices associated with that customer; and monetary value is defined as the total revenue over all their transactions.

The resulting customer-level table reveals the expected heavy-tailed structure. Histograms of recency, frequency, and monetary value show that most customers have made only a small number of purchases and have modest lifetime value, whereas a small minority are responsible for a disproportionate share of revenue. Rather than trimming outliers, which would effectively remove the most important customers from the analysis, all customers are retained, and skewness is addressed at the modeling stage.

To apply K-Means clustering, the three RFM features are standardized using scikit-learn’s StandardScaler. This removes differences in scale between the features so that K-Means is not dominated by the monetary dimension. The notebook then performs a model selection exercise over the number of clusters, k , ranging from two to fifteen. For each k , K-Means is fit to the standardized RFM data, inertia (within-cluster sum of squares) is recorded, and the silhouette score is computed to assess separation quality. The inertia curve exhibits a clear elbow around k equal to

four, and the silhouette scores peak in the same range, suggesting that four clusters strike a good balance between detail and interpretability.

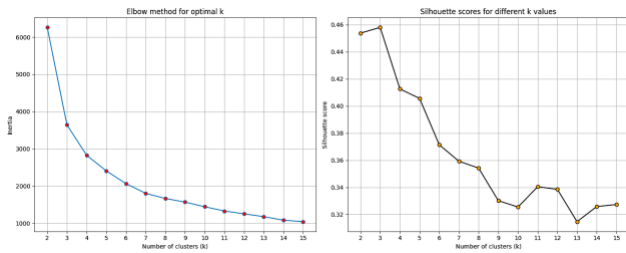


Figure 2: Elbow curve (left) and silhouette scores (right) for K-Means on standardized RFM features, showing that provides a good trade-off between model compactness and cluster separation.

A final K-Means model with k equal to four clusters is then fitted, and cluster labels are assigned to each customer. A three-dimensional scatter plot of standardized recency, frequency, and monetary value, with points colored by cluster, shows well-separated groups and provides an intuitive geometric picture of the segments. A cluster-wise summary of mean recency, frequency, and monetary value reveals that one cluster is characterized by very low recency, high frequency, and high monetary value; another has moderately low recency and solid frequency and spend; a third has fairly low recency but low frequency and spend; and the remaining cluster has very high recency and low engagement. These patterns motivate the following high-level interpretation: the most active, high-spending customers can be regarded as “Champions”; those with good but slightly lower engagement as “Loyal”; those who are recent but have not yet purchased much as “New” or “Potential”; and those whose last purchase dates back many months as “At-Risk” in terms of churn.

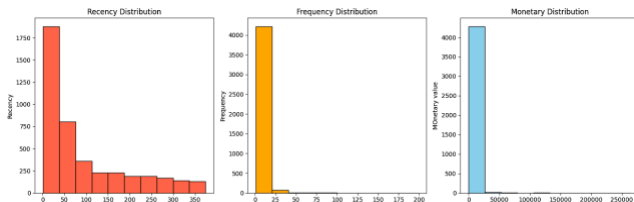


Figure 3: Histograms of Recency, Frequency, and Monetary features after aggregation to the customer level, highlighting the strong right-skew and heavy tails that motivate scaling and careful outlier handling before clustering.

2.3 Market Basket Analysis with Apriori

The second analytical component aims to uncover product combinations that frequently co-occur within the same basket. Starting from the cleaned transaction data, invoices are treated as baskets and products as items. A pivot operation groups the data by InvoiceNo and Description, sums quantities, and reshapes the result into a matrix in which rows correspond to invoices, columns correspond to product descriptions, and entries contain purchase quantities. This matrix is then binarized by mapping all

positive quantities to one and zero quantities to zero. The result is a sparse, binary invoice-product matrix suitable for frequent itemset mining.

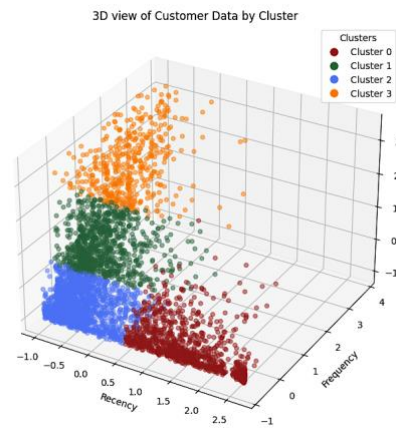


Figure 4: 3D scatter plot of standardized Recency, Frequency, and Monetary values colored by K-Means cluster, illustrating four well-separated behavioral customer segments.

Association rule mining is performed using the Apriori algorithm as implemented in mlxtend. The analysis explores several candidate settings for minimum support and maximum itemset length. In the final configuration used for reporting, the minimum support is set to 0.01 and the maximum itemset length to three. These parameters strike a compromise between computational tractability and interpretability, yielding a manageable number of frequent itemsets while still capturing non-trivial multi-item patterns. From the frequent itemsets, association rules are derived using lift as the primary interestingness metric. Rules are filtered to require lift greater than one, so that they represent positive associations, and a moderate confidence threshold (for example, at least 0.5) is applied to focus on rules that are both frequent and reliable.

The top rules sorted by lift are then visualized in a horizontal bar chart, where each bar corresponds to one rule and is annotated with the antecedent and consequent itemsets. While individual product names are specific to this retailer, a qualitative inspection of the rules reveals a clear pattern: many of the strongest associations occur among items that are conceptually part of the same collection, such as different color variants of a tableware line, or among complementary components of a multi-piece toy set. This structure is exactly what a retailer can exploit to design bundles (“complete the set”), to drive cross-selling (“customers who bought X also tended to buy Y”), and to coordinate inventory so that related items are stocked together.

min_support	Max itemset length	Frequent itemsets	Rules generated	Runtime (sec)	Observations
0.01	2	851	122	17.90	Many frequent itemsets and rules; top rules involve <i>Regency Tea Plate</i> colour variants and toy room sets.
0.01	3	901	251	18.91	Allowing three-item sets increases the rule count; new rules combine three variants (e.g., pink + roses ⇒ green).
0.01	4	903	264	18.55	Little change beyond three items, suggesting most frequent sets have ≤3 items.
0.02	2	232	20	1.22	Raising support sharply reduces itemsets and rules; strongest rules now involve <i>Regency Teacup</i> and <i>Saucer</i> colours.
0.02	3	233	25	1.22	Three-item rules emerge but are variants of the same teacup combinations.
0.02	4	233	25	1.16	No additional multi-item patterns beyond size 3.
0.03	2	87	0	0.28	At 3% support no rules meet the lift threshold; frequent itemsets exist but they are not predictive.
0.03	3	87	0	0.28	Same as above.
0.03	4	87	0	0.29	Same as above.

Figure 5: Summary of Apriori configurations showing how different combinations of minimum support and maximum itemset length affect the number of frequent itemsets, rules generated, runtime, and the types of patterns discovered.

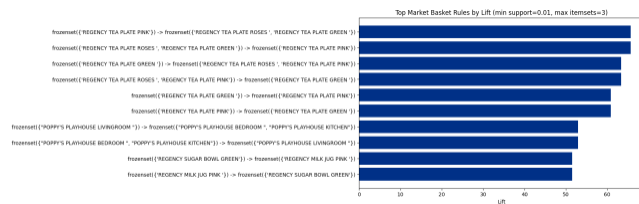


Figure 6: Top association rules ranked by lift for the final Apriori setting (min_support = 0.01, max itemset length = 3), highlighting strongly co-purchased product groups that correspond to tableware colour variants and complementary toy sets.

2.4 Sales Forecasting for Daily Revenue

The third component focuses on time-series forecasting for daily revenue in the United Kingdom. The country-level analysis begins by constructing a time series of daily revenue. For each transaction in the cleaned dataset, revenue is computed as Quantity multiplied by UnitPrice. The subset of transactions with Country equal to “United Kingdom” is extracted, and revenue is aggregated by invoice date at a daily frequency. The resulting series is resampled to a regular daily index, and missing days are filled implicitly with zeros where appropriate. A preliminary exploratory analysis plots the entire daily UK revenue series and reveals characteristic retail behavior: apparently stable baseline levels punctuated by weekly cycles and sharp increases in the final months of the year due to holiday shopping.

To formalize the weekly seasonality, the series is modelled as a seasonal ARIMA process. The daily revenue series for the UK is sorted by date and converted to a pandas Series with a daily frequency. A train–test split on 1 September 2011 is used: all observations up to and including that date form the training set, and the remaining dates through 9 December 2011 constitute the test set. This split ensures that the test portion covers the last quarter of 2011, including the main holiday season, which provides a stringent evaluation of forecast performance.

The initial baseline model is a SARIMA with non-seasonal order (1,0,1) and seasonal order (1,1,0,7), reflecting one autoregressive and one moving-average term at the daily level, seasonal differencing with period seven, and a weekly seasonal autoregressive component. This model is fitted on the training

data, and one-step-ahead forecasts are produced for the length of the test window. The baseline model achieves a test RMSE of approximately 28,060 and an MAE of approximately 19,016, providing a reference point for more flexible specifications.

Next, the notebook defines a small grid of candidate SARIMA orders around this baseline. The grid includes several combinations such as (5,0,1), (5,0,0), (1,0,5), (0,0,5), (2,0,2), (2,0,1), and (1,0,2), all with the same seasonal order (1,1,0,7). For each candidate model, the notebook fits the model on the training data, forecasts over the test period, and records the Akaike Information Criterion (AIC) together with RMSE and MAE on the test set. This systematic search reveals that the model with non-seasonal order (2,0,2) and seasonal order (1,1,0,7) offers the best trade-off. It yields an AIC of about 5467, an RMSE of approximately 25,108, and an MAE of approximately 15,639, improving substantially on both error metrics relative to the baseline.

The best SARIMA(2,0,2)(1,1,0,7) model is then refitted, and forecasts for the test period are plotted against the actual daily revenue, along with a 95% confidence interval derived from the state-space formulation. The plot shows that the model captures both the weekly oscillations and the general upward trend into the holiday season, although, unsurprisingly, the single highest holiday peak is underpredicted, as it behaves more like a rare event than a smoothly recurring seasonal fluctuation.

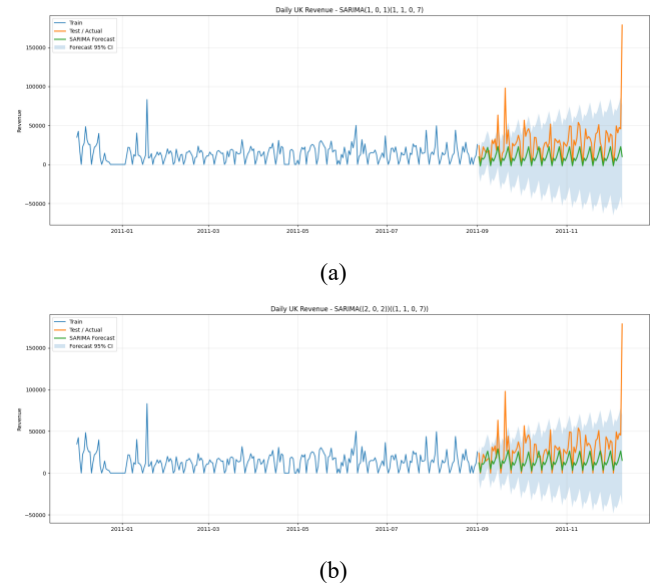


Figure 7: Daily UK revenue and SARIMA forecasts on the September–December 2011 test window. Panel (a) shows the baseline SARIMA(1, 0, 1)(1, 1, 0, 7) model, which captures weekly seasonality but tends to underestimate the level and variability during the holiday period. Panel (b) shows the tuned SARIMA(2, 0, 2)(1, 1, 0, 7) model, whose forecasts track the test-period dynamics more closely within the 95% prediction intervals, reflecting the improved fit reported in the error metrics.

To complement the SARIMA analysis, a Prophet model is also fitted. The UK daily revenue series is transformed into a data frame with columns `ds` (date) and `y` (revenue). The same train–test split on 1 September 2011 is applied. The Prophet configuration used here includes a trend component, weekly seasonality to account for the seven-day cycle, and neither daily nor yearly seasonality, since there is no intra-day resolution and the dataset covers only a little more than one calendar year. The resulting Prophet model attains a test RMSE of about 25,255 and an MAE of about 15,593, which is remarkably close to the tuned SARIMA model. This suggests that both methods are extracting essentially the same structure from the data: a smooth trend with a strong weekly seasonal pattern.

A second Prophet variant augments the model with an explicit “peak season” holiday component, defined by constructing a holiday data frame that marks all days in November and December for 2010 and 2011 as part of a peak season. This holiday-enhanced Prophet model produces a test RMSE of approximately 25,493 and an MAE of approximately 15,846. The slight deterioration in error relative to the simpler Prophet configuration indicates that the explicit holiday signal, as modeled here, does not add predictive power beyond the weekly seasonality and trend. The model still underestimates the single extreme holiday peak, underscoring how difficult it is to model one-off spikes without multiple years of training data.

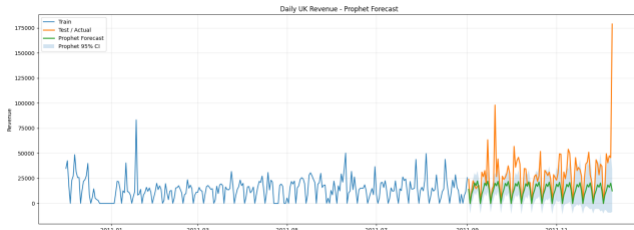


Figure 8: Daily UK revenue with Prophet forecasts on the September–December 2011 test window, showing predicted values and 95% confidence intervals overlaid on the training and actual test series.

2.5 Geographic Revenue Analysis

Although the primary focus of the forecasting module is the UK market, the notebook also constructs simple geographic summaries to place the UK in context. After filtering the cleaned data again to ensure positive quantities and unit prices, revenue and quantity are aggregated by country. A table of the top countries ranked by revenue shows that the United Kingdom overwhelmingly dominates, with a total revenue of approximately 6.62 million in the cleaned dataset. The next few countries, such as the Netherlands, Ireland, Germany, and France, contribute substantially less, both in revenue and in the number of invoices and customers.

To visualize this information, the analysis uses Plotly’s choropleth functionality to create a world map in which countries are colored by the log of their total revenue. To make the color scale more interpretable, the code also computes revenue in millions and the

log-transformed revenue for smoother gradients. Visual inspection of this map clearly highlights the UK as the primary market, with a scattering of secondary markets across Western Europe, Australia, and a few other regions. This map both confirms the decision to focus forecasting efforts on the UK and suggests where secondary country-specific analyses might be carried out in future work.

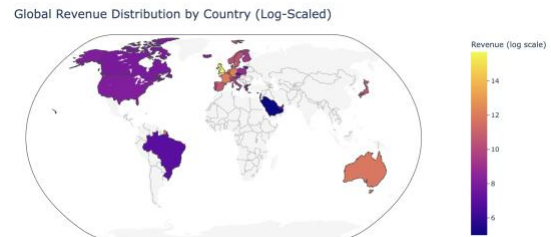


Figure 9: Global choropleth of total revenue by country (log-scaled), highlighting the dominance of the United Kingdom and other key markets while compressing smaller countries’ values onto a comparable color scale.

3 Empirical Experiments

The empirical evaluation mirrors the three main modules: segmentation, market basket analysis, and forecasting. All experiments are conducted within the same notebook on the cleaned dataset. For reproducibility and clarity, the notebook prints or stores key statistics, and all plots are generated directly from the code without manual editing.

For customer segmentation, the experimental design consists of first constructing the RFM features as described in Section 2.2, then standardizing them, and finally fitting K-Means models for different numbers of clusters. The evaluation uses two standard cluster validity indices: inertia (to identify elbows in the within-cluster sum of squares) and the silhouette coefficient (to assess average separation). Both metrics indicate that four clusters provide a good balance between compactness and separation. The four-cluster solution is then examined via the three-dimensional scatter of standardized recency, frequency, and monetary value. Cluster-wise averages show that one cluster has recency around 33 days, frequency near seven orders, and monetary value exceeding two thousand units; another has recency around 48 days and moderately high frequency and spend; a third has recency around 57 days but low frequency and spend; and the remaining cluster has recency of approximately 260 days and very low engagement. This validates the qualitative labels of Champions, Loyal, New/Potential, and At-Risk. The empirical outcome aligns with prior expectation that RFM captures essential behavioral heterogeneity in transactional settings.

For market basket analysis, the evaluation focuses primarily on the interpretability of the highest-lift association rules rather than on predictive accuracy per se. The combination of minimum support at the one-percent level and maximum itemset length of three yields a manageable rule set that still includes semantically

meaningful combinations. When the top rules are examined, one observes that antecedent and consequent itemsets tend to be drawn from the same product family or from complementary product categories. In many cases, rules reflect customers completing sets of coordinated items or buying accessories that are natural extensions of a focal product. The fact that lift values are substantially greater than one confirms that these co-occurrences are far more frequent than would be expected under independent purchasing, which gives a solid empirical justification for using these rules to drive bundle design and recommendation strategies.

The forecasting experiments are more quantitatively oriented. By explicitly reporting RMSE and MAE on a held-out test window, the notebook demonstrates that model refinement has tangible gains. The baseline SARIMA(1,0,1)(1,1,0,7) model is a sensible starting point: it encodes weekly seasonality and a simple autoregressive–moving-average structure and achieves an RMSE of about 28,060 and an MAE of about 19,016 on the Sep–Dec 2011 period. However, exploring a modest grid of alternative orders shows that the SARIMA(2,0,2)(1,1,0,7) model reduces RMSE by roughly 3,000 and MAE by about 3,400. These reductions are practically meaningful because they correspond to a sizeable improvement in average daily revenue prediction error. Prophet, with weekly seasonality and no explicit holidays, attains an RMSE of around 25,255 and an MAE of roughly 15,593, essentially matching the tuned SARIMA model. The Prophet variant with an explicit peak-season component performs slightly worse. Thus, the experiments demonstrate that both SARIMA and Prophet can be calibrated to yield similar performance, and that simple baselines can and should be improved upon by modest structural adjustments.

Overall, the empirical findings are consistent with domain expectations. RFM clustering naturally separates highly engaged, recent customers from stale and low-value ones. Association rules pick out within-collection and complementary-item patterns that any merchandiser would recognize as important. Time-series models capture weekly cycles and gradual ramps but struggle with isolated holiday spikes, which is expected given the short historical horizon and the rarity of extreme events.

4 Discussion

The project demonstrates that a carefully constructed data-mining pipeline can extract non-trivial insights from an ostensibly simple transaction table, but it also highlights several important limitations that should be acknowledged.

First, the dataset is fundamentally historical. It describes a particular retailer over a fixed period from 2009 to 2011, and there is no guarantee that the behavioral patterns uncovered—either in customer segments, product associations, or seasonal sales—will carry over unchanged to future years or to other firms. In particular, the strong holiday spike in late 2011 is modeled based on essentially one season, which is too little data to fully learn recurrent promotional structures or macroeconomic effects. The

models also have access only to transactional records; they do not observe marketing campaigns, promotions, price changes, website traffic, or macroeconomic variables. As a result, all systematic variation in revenue is implicitly attributed to autoregressive structure and seasonality, and the models cannot explicitly separate exogenous drivers from intrinsic dynamics.

Second, the segmentation module uses only RFM features. While RFM is a robust and interpretable framework, it ignores potentially rich information about product preferences, response to promotions, or browsing behavior. For example, two customers may have identical RFM profiles but very different category affinities or price sensitivities. An extension of the segmentation module could incorporate features such as the mix of product categories, discount usage, or sensitivity to seasonality, at the cost of moving beyond three-dimensional visualizability.

Third, the association rule mining uses relatively coarse hyperparameters. Raising minimum support reduces the number of rules and increases stability but may discard interesting niche patterns; lowering it produces a very large rule set dominated by noise. Likewise, limiting the maximum itemset length to three prevents combinatorial explosion but precludes the discovery of larger bundles. The project does not explore more advanced pattern mining techniques such as closed itemsets, sequence mining, or personalized recommendation models. It also does not embed any formal mechanism for selecting which rules to operationalize; that decision is left to human judgment.

Fourth, the forecasting component operates at the aggregate country level and assumes a univariate structure. Although forecasting aggregate daily revenue is useful for high-level planning, operational decisions often require finer granularity, such as category-level or product-level forecasts, and may benefit from multivariate models that incorporate exogenous variables like promotions, holidays, or macroeconomic indicators. The models also assume piecewise linear trends and stationary seasonal patterns, which may not hold in the presence of rapid structural changes in the business or the market.

Finally, the project is implemented in a notebook environment and is not packaged as a production system. There is no model deployment workflow, no batching or streaming inference pipeline, and no dashboard that would allow non-technical stakeholders to interact with the results. From a software engineering perspective, the analysis is best viewed as a research prototype.

Despite these limitations, the pipeline serves as a strong baseline and teaching example. It shows that a modest amount of modeling sophistication – R FM, K-Means, Apriori, SARIMA, Prophet – can substantially enhance the value of transactional data. It also illustrates standard modeling workflows: data cleaning and feature engineering, hyperparameter exploration, model comparison via train–test splits, and diagnostic plotting.

5 Conclusion and Future Work

This project has shown how to transform a raw e-commerce transaction log into a set of actionable insights through a cohesive data-mining pipeline. Starting from a noisy dataset with cancellations, administrative codes, and missing customer identifiers, careful preprocessing yields a clean set of 363,090 transactions that support both customer-level and time-series analyses. On this foundation, an RFM-based segmentation model exposes four behaviorally distinct customer clusters that align with intuitive marketing personas and suggest concrete retention and upsell strategies. Association rule mining on invoice–product baskets uncover product combinations that occur far more often than chance, validating the idea of designing bundles and cross-selling recommendations around these co-purchase patterns. Time-series forecasting for UK daily revenue, based on both SARIMA and Prophet, produces reasonably accurate short-term forecasts, improves significantly over simple baselines, and clarifies the structure of weekly and seasonal variation in demand.

The broader lesson is that even relatively simple, classical methods can deliver substantial value when applied carefully to transactional data. The models used here are transparent, interpretable, and well-understood theoretically, which facilitates communication with non-technical stakeholders and reduces the risk of overfitting or spurious correlations. At the same time, the results expose the limitations of solely relying on historical transactions. To fully exploit the potential of such data, future work could proceed in several directions.

One immediate extension is to enrich the segmentation with additional behavioral and product-mix features and to move from hard clustering to probabilistic models or latent class frameworks that can capture overlapping segments. A second direction is to refine the recommendation component by incorporating user- and item-based collaborative filtering or more modern representation-learning methods for market basket modeling, while still using the association rules as a transparent baseline. On the forecasting side, extending the analysis to category- and product-level series, incorporating exogenous regressors such as promotions and holidays, and experimenting with modern deep learning sequence models would all be natural next steps. Finally, from an engineering point of view, containerizing the pipeline, exposing its outputs through a simple API, and building a lightweight dashboard would move the work from a research notebook toward a deployable decision-support tool.

REFERENCES

- [1] D. Chen. “Online Retail II” (dataset). UCI Machine Learning Repository, 2012. DOI: 10.24432/CSCG6D.
- [2] Chen, D. (2012). Online Retail II [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/CSCG6D>
- [3] Chen, D., Sain, S. L., & Guo, K. (2012). Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining. *Journal of Database Marketing & Customer Strategy Management*, 19(3), 197–208. <https://doi.org/10.1057/dbm.2012.17>
- [4] Hyndman, R. J., & Khandakar, Y. (2008). Automatic time series forecasting: The forecast package for R. *Journal of Statistical Software*, 27(3), 1–22. <https://doi.org/10.18637/jss.v027.i03>
- [5] Christy, A. J., Umamakeswari, A., Priyatharsini, L., & Neyaa, A. (2021). RFM ranking – An effective approach to customer segmentation. *Journal of King Saud University – Computer and Information Sciences*, 33(10), 1251–1257. <https://doi.org/10.1016/j.jksuci.2018.09.004>